

Examiner reports

Q1.

In part (a), most correctly read the graph and then correctly used the coordinates to give the required P and V values.

The majority seemed well practised in the first step for part (b), writing the equation as $\log_{10} P = \log_{10} c + d \log_{10} V$. Many progressed to a correct solution easily and completed the question. Others became confused about what was a log and what was not or failed to handle the negative values and negative gradient. Those who tried to use point A and another point to generate simultaneous equations usually became lost at some point in the algebra.

In part (c), most remembered to include the units, relating the mathematical model back to the physical situation behind it.

Q2.

This question was well answered. Students were able to manipulate the given log expression confidently to reach the correct answer of $\log_a 12$. In a minority of cases errors

such as simplifying $\log_a 36 - \log_a 3$ to $\log_a 33$ and more commonly $\log_a 36 - \log_a 3$ to $\frac{\log_a 36}{\log_a 3}$ were seen.

Q3.

The majority of students scored full marks on (a), and the vast majority at least scored 1 or 2 marks for using a log law correctly.

(b) was less well answered with many students who identified that $\frac{3}{2}$ was not a valid solution failing to clearly explain why. Other students gave irrelevant comments criticising the setting out of the work rather than the mathematical validity.

Q7.

This question which tested the laws of logarithms caused problems for approximately half

the students. Although many correctly wrote $\log_a N - \log_a x$ as $\log_a \frac{N}{x}$, a common error

was to incorrectly write it as $\frac{\log_a N}{\log_a x}$. The majority of students who reached $\log_a \frac{N}{x} = \frac{3}{2}$ went on to present a full correct solution although others could not eliminate the logarithm correctly or failed to correctly change the subject of the resulting formula to write x in terms of N and a .

Q8.

In this question, which tested logarithms, most students scored the mark in part (a) but not all could see its use at a later stage in part (b). Not unexpectedly, part (b) proved discriminating, with a majority of students only picking up one of the six marks, normally

for writing $2\log(x + 7)$ as $\log(x + 7)^2$. The remaining students, who correctly applied a

further law of logarithms to reach the stage $\frac{(x+7)^2}{x+5} = 3$, either stopped at that point or went on to find the correct quadratic equation. In the latter case, most went on to solve the equation correctly but the required concluding statement 'only one value of x '. was not always given and so the final mark was lost. Some students who recognised what was being asked for frequently obtained the correct quadratic equation, showed that $b^2 - 4ac = 0$ and then stated 'only one value of x '. for full marks.

Q9.

This question, which tested the 'exponentials and logarithms' section of the specification, resulted in a good spread of marks. Although there were many correct sketches of $y =$

$\frac{1}{2^x}$ seen in solutions to part (a), there were almost as many that were sketches similar to that of $y = 2^x$. Another common incorrect sketch was one which was obtained by reflecting $y = 2^x$ in the x -axis.

Part (b) was generally answered well, although some candidates failed to pick up the final mark because they either left their final answer in the form $-x = 0.322$ or did not give the value of x to at least the required degree of accuracy. Of those who were less successful, some displayed poor algebraic skills or believed incorrectly that cross multiplying led to an equation of the form $10^x = 4$.

Part (c) proved to be demanding, with most candidates failing to obtain the correct expression for y in terms of a and b , although the majority of candidates were awarded partial credit for applying log laws correctly. A typical misconception is illustrated by $2\log_a b + 3\log_a y = 5\log_a by$. Those candidates who were the most successful usually applied no more than one log law (or used $\log_a a = 1$) per line of working to reach, for example, $\log_a b^2 y = 1$ and then stated $b^2 y = a$ before completing by rearranging to obtain $y = ab^{-2}$.

Q10.

Approximately half the candidates presented a fully correct solution to part (a). Although most of the other candidates scored a mark for applying a correct logarithmic law, normally writing

$2\log_k x - \log_k 5 = 1$ as $\log_k x^2 - \log_k 5 = 1$ as $\log_k 5 = 1$, a significant proportion of these candidates either

made the error illustrated by ' $\log_k 5 = 1$ as $\log_k x^2 - \log_k 5 = 1$ as $\log_k 5 = 1$ so $\frac{\log_k x^2}{\log_k 5} = 1$ ', or made no further progress.

Another common error, worthy of note, involves the lack of appreciation that the coefficient 2 in the expression $2\log_k 5 = 1$ as $\log_k x - \log_k 5 = 1$ as $\log_k 5$ has no affect on the term $\log_k 5 = 1$ as $\log_k 5$. It was not uncommon to see

$2\log_k 5 = 1$ as $\log_k x - \log_k 5 = 1$ as $\log_k 5$ replaced by $2\log_k \left(\frac{x}{5}\right)$.

Part (b) proved to be more challenging than part (a). A majority of candidates scored one mark

for correctly rewriting at least one of the given logarithmic equations in a non-logarithmic form

and many of these candidates were able to eliminate a to obtain the equation $y = (4^{b+2})^{\frac{3}{2}}$ and so

scored the second mark. However, relatively few candidates were able to show that the right-hand side could be written as $2^3 k 5 = 1$ as $\log^{b+6}$. The most common error was to

write $(4^{b+2})^{\frac{3}{2}}$ as $2^{\frac{3}{4}(b+2)}$.

Q11.

A very large percentage of the candidates who attempted this question found the correct value for the sum to infinity in part (a).

In part (b) the majority of candidates got the correct 6^{th} term but a large proportion felt that evaluating using a calculator (rather than changing to powers of 2 and 3) was an appropriate

way of showing that $12\left(\frac{3}{8}\right)^5$ and $\frac{3^6}{2^{13}}$ were equal. Only a very small minority of candidates confused the 6^{th} term with the sum to 6 terms.

In part (c)(i), many candidates could state the general formula for the n^{th} term but a significant minority failed to substitute the necessary values. A number of candidates, without proof, used either their answer to part (b) or the given answer in part (c)(ii) to write down their answer in terms of powers of 2 and 3. There were relatively few fully correct solutions to part (c)(ii) though quite a few did pick up some marks.

Starting with their u_n from part (c)(i) a common error was to think that $\log ab$ was $\log a \times \log b$ rather than $\log a + \log b$ or to take the power to the front of it all. This meant that the correct use of a further log law was often the only mark that was gained. Many who correctly applied two laws of logarithms then struggled to write their expression in terms of $\log 2$ and $\log 3$ only. Although rare, there were some excellent solutions seen which differed from the candidates' normal approach.

One such approach, for full marks, is illustrated by ' $u_n = 12 \times \left(\frac{3}{8}\right)^{n-1} = 32 \times \frac{3^n}{8^n} = \frac{2^5 \times 3^n}{2^{3n}}$;
 $u_n = 3^n \times 2^{5-3n}$, $\log_a u_n = \log_a 3^n + \log_a 2^{5-3n} = n \log_a 3 + (5 - 3n) \log_a 2 = n \log_a 3 - (3n - 5) \log_a 2$ '

Credit was given to those candidates who presented the last two stages of the above solution but, without the first of the three stages at some point in their solution to the question, only half marks could be awarded.

Q12.

Part (a) was well done.

In part (b) the majority recognised what was required and obtained the relevant quadratic even if their notation was sometimes a bit suspect. A few used a substitution for $\ln x$ which helped, but those who used x instead of, say Y , occasionally forgot to change back to earn the final marks. The occasional candidate changed $3\ln x$ to $\ln(x^3)$ which was not helpful, and after multiplication by $\ln x$ there were a few $(3\ln x)^2$ terms seen.

Q13.

The candidates were mostly familiar with the techniques associated with logarithms, but the relationships between a and c , and between b and m , were often left in logarithmic form rather than in the way specified in the question, which involved exponential forms. These exponentials could perhaps have pointed candidates in the right direction when they came to answering part (c), where antilogarithms were needed for both answers. The linear graphs in part (b) were mostly plotted and drawn accurately, though some candidates misused the scale on the vertical axis.

Q14.

Many candidates obtained the correct values for x in parts (a)(i) and (a)(ii). The more able candidates often presented a full correct solution when solving the logarithmic equation in part (b) but a common error, seen in solutions from less able candidates, was to assume that $\log(n - 4) = \log n - \log 4$, which frequently resulted in a single value for n even though the question asked for the possible values of n .

Q15.

Parts (a) and (b) gave the opportunity for all to pick up marks, with the majority giving the two decimal place answer required in part (b). Many candidates scored just one of the two marks in part (c). The translation approach was the more popular, but the fact that the

vector needed to be $\begin{bmatrix} 3 \\ \frac{4}{3} \\ 0 \end{bmatrix}$ was too subtle for most, as was the scale factor of $\frac{1}{8}$ for those using a stretch approach. Similarly in part (d), needing $4(x - 1)$ rather than $4x - 1$ defeated the majority. However, this caused no undermining of confidence as many answered part e(i) well, with sufficient justification to support the given answer.

It should be noted, however, that answers which simply started with $k = \frac{8 \times 5}{4}$ with no evidence of using correct log laws could not earn credit. In part (e)(ii) most candidates

knew they had to solve ' $\frac{5}{4} = 2^{4x-3}$ ' and started by taking logs, but then resorted to calculators rather than establish an exact result (with numbers hinted at in the previous part). The best solution, given by some, was to multiply both sides by ' 2^3 ' and then take logs, to the base 10, of ' $10 = 2^{4x}$ ', which led directly to the result.

Q16.

Most candidates were able to score marks in part (a), although the mark in part (a)(iii) was not awarded for answers left as ' $\log_a 5^3$ '. In part (b), candidates generally scored the method marks but frequently failed to pick up the accuracy mark. The most common

wrong answer was '1.66', not always simply from a premature approximation.

Part (c), as expected, proved to be beyond the ability of the average candidate; some did score a mark for ' $p = 2^m$ ', but this was often followed by ' $pq = 2^m \times 8^n = 16^{mn}$ '. Better candidates reached ' $pq = 2^m \times 2^{3n}$ ', but a significant number of these candidates then simplified incorrectly to reach the wrong answer ' $pq = 2^{3mn}$ '.

Q17.

The work required to answer parts (a)(i) and (a)(ii) was well understood but a significant number of candidates did not give an explicit value for p or an explicit value for x . Candidates at this level should not be leaving their answers in an embedded form, for example, in part (a)(ii) answers were sometimes left as ' $52 \times 0.75 = 125$ '.

Part (b) was answered better than the corresponding work in previous years and there continued to be an improvement in knowledge and use of the laws of logarithms as seen in solutions to part (c) where many candidates were able to score at least half marks by correctly applying two laws of logarithms. Only more able candidates either used $\log_a a = 1$

or applied $\log_a \frac{x}{k} = -1 \Rightarrow \frac{x}{k} = a^{-1}$ towards expressing x in terms of a correctly.

Q18.

Many candidates gave a correct solution for part (a) but only a minority gave full correct solutions to part (b). Average grade candidates in general obtained the method marks for applying two logarithmic laws correctly but only the more able candidates could deal with the constant 1 correctly. Instead of writing 1 as $\log_a a$, the common wrong approach was to write $\log_a 36 + 1$ as $\log_a 37$.

Q19.

Part (a)(i) was well answered.

Candidates who saw the connection with part (a)(i), usually made a very good attempt at part (a)(ii), often with complete success. Limits did cause a problem where candidates had not bracketed $2x^2 - 8x + 3$, and some candidates did not leave the answer in the required

form but left their answer as $\frac{1}{4} \ln 9$.

Part (b) was well answered by candidates who were able to write the integral in terms of u successfully. Many candidates lost a mark through omission of either du or c . A disturbing

number of candidates thought that $\frac{1}{3} \times \frac{1}{3} = \frac{1}{6}$. Some clearly able candidates lost the last mark by spoiling an otherwise acceptable answer.

Q20.

This question, surprisingly, was not done well. Part (a) was intended as a straight forward partial fractions question, although some candidates did not see it that way. Many apparently guessed at the value of k , 3 being a common wrong answer. Some candidates

replaced the k with the conventional A and B , and then demonstrated that $A = B$. Many of these, having arrived correctly at $6A = 3$, then deduced that $A = 2$. This 2, or other values,

often became $\frac{1}{2}$ in part (b) without any supporting evidence and so lost further marks. Very few candidates in this situation seemed to have checked for a mistake, despite the $\frac{1}{2}$ being given in part (b).

In part (b), most candidates picked up a mark for recognising a log integral, although the coefficients were often incorrect, particularly the sign in integrating the $\frac{1}{3-x}$ term. However, most candidates could evaluate their log expression correctly using the laws of logs.

Q21.

This question, which tested the basic definition of a logarithm as well as testing the laws of logarithms, was answered much better than in previous series. The most common error

was in part (b), where $\log_a 30 - \log_a 15 = \frac{\log_a 30}{\log_a 15}$ was seen quite often.

Q22.

Most solutions to part (a) were correct but some weaker candidates resorted to using 10 for a or just stated " $x = 8/3$ ". The majority of candidates gained credit for applying a correct law of logarithms in part (b) but many then made the error of writing $\log_a 6^3 - \log_a 8$ as $\frac{\log_a 6^3}{\log_a 8}$

Hence, full credit was not awarded to those candidates who went directly from $\log_a 6^3 - \log_a 8$ to the printed answer: $\log_a 27$. In part (c)(i), a significant minority failed to introduce

p and left their final answer as " $y = \log_{10} \left(\frac{27}{8} \right)$ ". The final part of this question was frequently not attempted. Although some very good solutions were presented, it was surprising to see some otherwise able candidates quoting wrong formulae; " $\text{gradient} = (x_2 - x_1) \div (y_2 - y_1)$ " was the most common of these.

Q23.

Part (a) of this question was well answered, most candidates being familiar with the equation $y = ax^b$ and the technique needed to convert it into linear form. Some candidates seemed less happy with part (b), but most managed to show enough knowledge to score well here. Errors often arose from confusion between the intercept 1 on the vertical axis and the corresponding value of y , which required the taking of an antilogarithm.

Q24.

Many candidates correctly used the laws of logarithms to write $\log_a 3(2n - 1)$ but then made an error in expanding the brackets either before or after eliminating the logarithms. A surprising number of candidates reached the correct equation ' $n = 6n - 3$ ' but were then unable to solve it correctly. Some candidates rearranged the given equation to have 0 on one side but these candidates often made an error in their subsequent work when trying to eliminate the logarithm. Weaker candidates started by writing ' $\log_a n = \log_a 3 + \log_a 2n - \log_a 1$ ' and so scored no marks for part (a).

Those candidates who attempted part (b)(i) generally obtained the correct expression, although the usual wrong answers such as 3^a or $\log 3_a$ were still seen.

In part (b)(ii) a significant number of above average candidates scored the first two marks for ' $\log_a (y/8) = 4$ ', but, surprisingly, some candidates who had answered part (b)(i) correctly could not eliminate the logarithm to find y in terms of a . Those candidates who tried to multiply by x or $\log x$ at an earlier stage often got themselves into a mass of incorrect algebraic work.

Q25.

Although this question on logarithms was answered better than corresponding questions in summer 2005, the topic continues to be one which candidates find difficult. Many candidates gained credit for writing ' $2\log n = \log n^2$ ' but then often made the error ' $\log(5n - 24) = \log 5n - \log 24$ '. In part (b) it was disappointing to find so many candidates unable to solve the quadratic equation. It was not uncommon to find solutions which involved the wrong factors of the form $(n \pm 24)(n \pm 4)$. In general, only the better candidates gave a fully correct solution for both parts of this question.

Q26.

This question on logarithms was answered better than questions on the same topic last year although this area of the specification continues to cause problems for candidates. Most candidates appreciated that $21\log_a 6$ is equal to $\log_a 6^2$ but then a significant number

of candidates made the error of equating $\log_a 36 - \log_a 3$ to $\frac{\log_a 36}{\log_a 3}$. Examiners expected to see an intermediate stage between ' $\log_a x = \log_a 36 - \log_a 3$ ' and the printed answer. In part (b), many candidates were able to write the left-hand side of the given equation as a single logarithm but then could not move out of logarithms. A significant number of answers did not involve the constant a or gave the answer as ' $5y = 7^a$ '. Although the question asked for y in terms of a , examiners condoned a final answer left as $5y = a^7$.

Q27.

In part (a), some candidates did not provide sufficient steps to justify stating the printed

value for x . The most common error was illustrated by " $\log_a 216 - \log_a 8 = \frac{\log_a 216}{\log_a 8}$ ". Most candidates gained significant credit in part (b), although part (b)(iv) did provide some challenge.

Q28.

This question on geometric series and logarithms proved to be very difficult for many candidates, with a significant number of the weaker candidates unable to quote correct relevant formulae from the booklet. Transforming the first sentence of the question into a correct equation was beyond many, but partial credit was usually gained by using the printed value for r . This part of the question seemed to show up a lack of understanding of algebraic manipulation. Rearranging ' $4a(1 - r) = a$ ' to ' $r = 3/4$ ' proved to be more difficult than anticipated. In part (b) it was very evident that a significant number of candidates did not understand the meaning of the letter ' a ' in the formula for the sum to n terms of a geometric series. To their credit, some overcame this difficulty by working out S_1 as

$$\frac{a(1 - r^1)}{(1 - r)} = a \text{ and then stating 'so first term is } a'.$$

Those who used the correct formula from the booklet normally obtained the method mark but a significant proportion of these candidates could not evaluate the expression correctly on their calculators. Those candidates who resorted to listing the 10 terms and then adding them rarely obtained the correct four decimal place answer and clearly used up valuable time in the process.

Although some excellent solutions to all parts of (c) were seen, relatively a few candidates went beyond part (c)(i). Many were able to quote the correct expression for u_n from the formulae booklet. For u_{2n} a common incorrect answer was $2ar^{n-1}$. It was very common to find candidates using ar^{n-1} to mean $(ar)^{n-1}$ which, although not penalised in part (c)(i) lead to mistakes by candidates in reaching the printed answer in part (c)(ii).

Those candidates who recognised the left-hand side of part (c)(ii) as $\log_{10} \left(\frac{2a_n}{2a_{2n}} \right)$ and then cancelled the a 's were far more successful than those who did not use the law of logarithms before substituting. It was pleasing to see some candidates, having abandoned their solution to part (c)(ii), using the printed answer with $n = 100$ to gain credit in the final part of the question. For full credit, examiners expected to see some evaluation of 100

$\log_{10} \left(\frac{4}{3} \right)$ before writing the printed answer, for example ' $\approx 12.49 \approx 12.5$ ' was acceptable.

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Q29.

Many candidates were able to give the correct expansion in part (a), normally by applying Pascal's triangle. Many realised that x should be replaced by reaching the exact printed answer. Candidates should be aware that approximate decimal values for each of 4

$\sqrt{5}$ and $4(\sqrt{5})^3$ cannot then be added and the result used to equate to the exact surd part of the printed answer. The final part of the question proved difficult, although, again, some elegant solutions were presented. Those who wrote ' $\log_2(1 + \sqrt{5})^4 = \log_2(56 + 24\sqrt{5}) = \log_2[8(7 + 3\sqrt{5})]$ ' usually went on to score all three marks.